

LLM Triumvirate as Physical Plausibility Judge: A Benchmark for Physics-Aware Test Verdict Generation

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Three small local LLMs (Google, MistralAI, Meta) judge the physical plausibility of state transitions through an explicit physics prior, with no GPU and no training: **8/9 correct verdicts**, 7/9 unanimous, and one failure on a static levitation scenario.

1. Motivation & Problem

Judging whether a physical transition is plausible has no ground-truth oracle: this is a classic instance of the **test oracle problem** in software testing.

- World Models (e.g. JEPA-based LeWorldModel, Maes et al., 2026) learn physical plausibility from raw pixels, but require a GPU, thousands of training episodes, and domain-specific retraining.
- Research question: can small, general-purpose LLMs judge physical plausibility directly from text, with no training and no GPU?

Approach in one line

Three small LLMs vote independently on the same state-transition, using an explicit physics prior injected via system prompt.

The exact system prompt (verbatim excerpt)

Evaluate the physical plausibility of this state transition. You are a physical plausibility judge. You reason about state transitions in a simple 2D or 3D environment.

Physical laws you apply: [see the six laws in Panel 4, energy, continuity, gravity, friction, inertia, momentum]

State format:
position: [x, y] or [x, y, z] in meters
velocity: [vx, vy] or [vx, vy, vz] in m/s
on_ground: boolean
context: optional - gravity, environment, or special conditions

Action format:
force: [fx, fy] or [fx, fy, fz] in Newtons
dt: timestep in seconds

Your response must be strict JSON, nothing else - see Panel 4 for the output schema.
No text before or after the JSON.

3. The LLM Triumvirate

Three independent judges, majority vote, three different model families, a consensus pseudo-oracle. Each judge gets the same input and prior, and returns the same JSON verdict.

Input triplet
state_t, action,
state_t+1, context

Physics prior
system prompt,
6 laws, g configurable

Gemma-3n
Google, 6.9B

Mistral-7B
MistralAI, 7B

Llama-3.2-3B
Meta, 3B

Majority vote
3/3 unanimous:
robust verdict
2/3 split:
flag for review

Retained models (all local, CPU-only, GGUF Q4_K_M)

Role	Model	Provider	Size
Primary judge	Gemma-3n	Google DeepMind	6.9B / 4.2GB
Analytical judge	Mistral-7B-Instruct-v0.3	MistralAI	7B / 4.1GB
Lightweight judge	Llama-3.2-3B-Instruct	Meta AI	3B / 2.0GB

Why three?

- Reduces variance from non-deterministic small-LLM behaviour
- Different architectures avoid a single training-regime bias
- Unanimous vs. split verdict gives a built-in confidence signal

What do "primary / analytical / lightweight" mean?

The labels reflect average inference time per scenario (Gemma ~50s, Mistral ~70s, Llama ~30s), not authority. **All three votes count equally**: the verdict is a plain majority vote.

A Kantian reading. The six laws are the universal, each scenario is the particular, and each judge performs a determinant judgment (Kant, 1781, 1790). Equal votes echo the *sensus communis*: a verdict is valid through agreement across judges, not through any single one.

5. Benchmark & Results

9 VoE scenarios, expected vs. observed verdict

ID	Scenario	Expected	Outcome
S1	Inertial push (2D)	Plausible	Correct, split (2/3)
S2	Discontinuous jump (2D)	Implausible	Correct, unanimous
S3	Spontaneous acceleration	Implausible	Correct, unanimous
S4a	Static levitation, Earth	Implausible	Failed , split (2/3 Plausible)
S4b	Static levitation, ISS (g=0)	Plausible	Correct, unanimous
S5	Free fall, Earth	Plausible	Correct, unanimous
S6	Free fall, Mars	Plausible	Correct, unanimous
S7	Inertial push (3D)	Plausible	Correct, unanimous
S8	Discontinuous jump (3D)	Implausible	Correct, unanimous

8/9 correct verdicts · 7/9 unanimous

Sustainable AI

- No GPU: a 2021 consumer laptop (Dell Latitude 5420, Intel i5, 24GB RAM)
- About 10GB of open-source GGUF weights, served locally with LM Studio
- About 25 minutes for the full benchmark (9 scenarios, 3 judges)

2. Grounding: the VoE Paradigm

The Violation-of-Expectation (VoE) paradigm comes from developmental psychology: infants look longer at events that violate their physical expectations.

- Canonical case (Needham & Baillargeon, 1993): 4.5-month-olds expect a released box to fall, they show surprise when it stays stationary in mid-air.
- VoE has since been adopted to evaluate AI systems (World Models) on physical common sense.
- This work asks the same VoE-style question of LLMs instead of pixel-based models or infants.

The symmetric pair S4a / S4b: raw results

Same state, a cube floating in mid-air with zero force applied. Only the context changes: Earth gravity (S4a) vs. ISS microgravity, g = 0 (S4b). Each cell: vote, score, time.

Judge	S4a, Earth (expected: implausible)	S4b, ISS (expected: plausible)
Gemma-3n	Plausible, 0.8, 48.4s	Plausible, 1.0, 59.6s
Llama-3.2-3B	Plausible, 1.0, 26.7s	Plausible, 1.0, 34.1s
Mistral-7B	Implausible, 0.0, 66.4s (violation: gravity)	Plausible, 1.0, 78.9s
Verdict	PLAUSIBLE, 2/3, split, avg 0.6: error	PLAUSIBLE, 3/3, unanimous, avg 1.0: correct

What the judges said on S4a (Earth)

- Gemma-3n** accepts the float: "no change in position or velocity, which is physically plausible for a floating object".
- Llama-3.2-3B** only checks inertia: "consistent with inertia and no external forces acting on it".
- Mistral-7B** applies gravity: "floating without any force being applied to counteract gravity, which contradicts the law of gravity".

On S4b (ISS), all three agree that in microgravity no force means no change of state.

4. Physics Prior & Prompt

Each scenario is a state-transition triplet (state_t, action, state_{t+1}), the same input format a pixel-based world model would receive.

Six physical laws in the system prompt

- Energy conservation** ("an object cannot accelerate without an applied force")
- Spatial continuity** ("an object cannot teleport, position varies continuously")
- Gravity** ("g = 9.81 m/s² on Earth unless another value is specified in context")
- Friction** ("objects on the ground decelerate without a driving force")
- Inertia** ("a moving object continues unless opposed by a force")
- Collision momentum** ("post-collision velocity respects conservation of momentum")

Required structured verdict (JSON)

plausible (bool), *score* (0.0–1.0), *violations* (list), *reasoning* (2–3 sentences), a test-report-like output.

```
{  "plausible": false,  "score": 0.0,  "violations": ["spatial continuity"],  "reasoning": "The block moves 2.8 m in 0.1 s with no applied force, implying 28 m/s, a discontinuous jump."}
```

Test engineering view

Testing concept	In this work
Test case (IEEE 829, ISO/IEC 29119)	One scenario: state_t, context, action, expected verdict
Oracle specification	System prompt: six physical laws, configurable context
Test report	JSON verdict: plausible, score, violations, reasoning
Consensus pseudo-oracle	Majority vote of three LLM judges
Metamorphic relation	S4a / S4b: same state, Earth vs. ISS context, the verdict must flip

6. Discussion & Conclusion

The S4a failure

Two of three models cannot override a strong text-based prior that "floating objects" are often legitimate (shelves, aerial views, zero-gravity fiction), unlike infants or pixel-based models, the LLMs have no visual grounding to disambiguate a static scene.

Limitations

- Nine synthetic scenarios designed by a single researcher
- Correct verdict does not guarantee correct reasoning, reasoning must still be inspected manually
- Not yet tested on noisy, real-world sensor data
- Verdicts stable at temperature 0.0, S4a included; non-zero temperature runs are future work

Take-away

A small, GPU-free LLM triumvirate can serve as a reproducible, low-cost test oracle for physical plausibility on dynamic violations, with a known blind spot on static, visually-ambiguous scenes, pointing to a hybrid LLM plus world-model architecture as future work. The single failure on the static gravity scenario highlights an important limitation of relying on textual physical priors: without visual or sensorimotor grounding, a strong language prior can override an explicit physical constraint on statically ambiguous configurations.